The Fragile Edge: Chaos Engineering for Reliable IoT

Chaos engineering is a great way of detecting possible failures in IoT devices. This technology has evolved well for testing cloud failure, but open source communities are still working towards building an efficient chaos engineering toolkit for testing IoT devices.

Billions of devices around the world today are connected to the internet, performing a variety of tasks in real time. According to *IoT Analytics*, the total number of IoT devices worldwide is expected to exceed 21 billion by 2025.

While the number of IoT devices continues to grow, they remain vulnerable to malfunctions. Most IoT devices use basic microcontrollers with only a few megabytes of memory. A smart thermostat worth a few hundred rupees has very little processing power. This limits it to handling only a few tasks at a time. As a result, these devices often fail when they encounter unexpected operating conditions.

Network instability makes IoT systems even more difficult to manage. Edge devices often rely on Wi-Fi, Bluetooth, cellular networks or LoRaWAN -- a low power wireless technology that connects devices over long distances while using very little power. These networks are not always reliable, particularly in remote areas or congested cities. For instance, a smart doorbell may momentarily lose connectivity. In large industrial setups, millions of devices can experience disconnections, lost data packets and delays in communication. These are network disruptions. Studies show that in real world IoT systems, up to 5–15% of data packets can be lost. Latency can vary from milliseconds to several seconds depending on conditions. These factors make the smooth operation of IoT systems challenging.

The use of many different hardware components makes it harder to keep IoT systems stable. The Internet of Things includes tens of thousands of different chips, operating systems and communication protocols. Some devices run Linux, while others use tiny real-time operating systems to manage their functions. The storage in these devices mainly uses flash memory and volatile memory (RAM) for processing. As there are so many different types of devices, it is impossible to create a single solution to ensure stability and performance for all of them. Real world events demonstrate how easily these systems can fail. For example, a factory may stop production because its sensor network stops sending data due to firmware problems.

The combination of these factors causes edge failures, which have become increasingly common. Batteries may run out sooner than anticipated, devices may freeze, connections may malfunction, and sensors may provide inaccurate readings. Serious issues arise when transportation networks, healthcare systems and industrial operations malfunction. They can cause delayed medical alerts and stop factory production. Smart home system failures also create real problems, such as smart locks that do not work and security cameras that stop operating. This is where chaos engineering helps solve the problem. Through controlled failure tests, engineers can predict when systems may fail. This enables them to create systems that function normally even in the event of failure.